

Human in the Loop Learning through Grounded Interaction in Games

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Abstract

Over the past few years, methods for learning from interaction have become a crucial paradigm in Artificial Intelligence, and we are now witnessing a growing interest in learning from *grounded* interaction, in particular through *dialogue games*. In the Dagstuhl Perspectives Workshop 24492, “Human-in-the-Loop Learning through Grounded Interaction in Games”, we discussed these new developments, and identified a few crucial directions for this research. These directions were considering agent behavior in complex interaction; ensuring that games properly tested all aspects of an agent’s cognitive and communicative ability; considering the types of grounding required at all levels of interaction; and developing new training methods that could fully leverage these richer types of context and communication.

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1 Executive Summary

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Background and Motivation

Over the past few years, there has been a decisive move in Artificial Intelligence (AI) towards human-centered intelligence and towards AI models that can learn through interaction. An important reason for this shift has been the appearance of the latest generation of Large

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Human in the Loop Learning through Grounded Interaction in Games

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Language Models such as InstructGPT, ChatGPT, BARD, or Lamda-2 [13, 12, 17] capable of a step-increase in performance. A good part of the success of these models is due to the adoption of training regimes involving a combination of supervised learning and learning from interaction with humans, such as Reinforcement Learning Through Human Feedback [2, 13]. And particularly the most recent among such models, such as GPT-4, are not simply language models, but are trained with multimodal data and are capable of producing output in different modalities as well. However, these models still suffer from a number of widely discussed issues, such as hallucinations.

In parallel, there has also been substantial progress on grounded interaction – developing models aware of the situation in which they operate (a physical world in the case of robots, a virtual world in the case of artificial agents) and able to, e.g., understand / produce references to this situation [4, 7, 8, 1, 16] perhaps through negotiation [3]. However, the communication between the interactive learning and grounded interaction communities is still limited [10].

One domain considered particularly promising to study learning through grounded interaction with human agents is virtual world games: games in which conversational agents impersonating characters can learn to perform tasks, or improve their communicative ability, by interacting with human players in platforms such as Minecraft or Light [6, 19, 11, 15, 9, 24]. Games have been shown to be a promising platform for collecting data from thousands of players [20, 23]; virtual worlds approach the complexity of the real world; and virtual agents operating in such virtual worlds need to be able to develop a variety of interactional skills to be perceived as “real” [14].

This Dagstuhl Perspectives Workshop aimed, first of all, to bring together the communities working on the related areas of learning through interaction, (conversational) agents in games, dialogue and interaction, and collecting judgments from crowds through games, to make each community aware of the most recent developments in the other areas. We also intended to discuss current challenges, and whether advances in one area (e.g., grounded interaction) can benefit other areas (e.g., interactive learning).

Directions Identified and Discussed

The workshop involved extensive discussions between researchers working in all the fields that contribute to the research area. After in-depth presentations of:

- The State of the Art (SOTA) in The Grounded and Communication Task Performance Abilities of (Embodied/Multimodal, Conversational) AI Agents (by group led by Hockenmaier)
- The Games and Multimodal Platforms Useful for Conversational AI Agents (group lead: Bernardi) and
- Current Approaches to Human (and Artificial Agent)-in-the-Loop Learning for AI Agents (group lead: Suglia),

and presentation of some of the most recent relevant research by the participants, we identified a few research directions particularly worth discussing in depth, and formed working groups around them. These included:

■ **Complex Interaction.**

A common assumption in many computational models is that dialogue consists of a linear sequence of turns in which two agents alternately exchange information. Each turn is assumed to depend only on the last turn of the other participant. However, human conversation requires more complex forms of interaction spanning multiple turns to solve real-world tasks.

Complexity occurs for several reasons: Dialogue is done by multiple people. They start from different information states, have different perspectives, and cannot see what is in each other’s minds. They have a social relationship that they have to manage. Their interaction happens in real time, across multiple modalities, in the presence not only of various kinds of noise but of fundamental asymmetries in what the participants can perceive, know and understand.

To successfully overcome these asymmetries and solve tasks through such interactions, the interaction scheme needs to offer a number of functions (see below). Among humans, these are exemplified by a variety of phenomena that depend not just on sequential information exchange but on more complex structures, with richer models of the local and global interaction context. It is not clear to what degree current LLM-based models of dialogue can cope with them, and how much this limits their ability to collaborate efficiently with humans.

The working group on Complex Interaction reviewed some of these complex interaction phenomena, gave pointers to the literature, and discussed ways in which future interactive systems might handle them.

■ **Game Design for Grounded Interaction.**

Existing games and platforms used to evaluate and develop conversational agents are extremely diverse in their setting, goals, and complexities, and they are being developed in different subfields of AI, NLP, and computational linguistics [5]. Furthermore, within these subfields, games are designed for different purposes and, to some extent, classified using different taxonomies.

The explosion and diversity of games raise new research questions for these communities that we suggest should be explored in future research:

- Q1: How can games and game benchmarks be designed more systematically, such that they lead to a deeper understanding of games and the skills that games are testing? How do we generalize skills and agents’ abilities across games?
- Q2: What role does the complexity of the game have? And how do we measure it?
- Q3: How do we evaluate agents within and across games? In particular, how do we evaluate whether the skills trained / tested with a game transfer to real world applications?

■ **Perspectives for Language Learning from Human Interaction.**

Most of the recent AI breakthroughs have been in non-interactive settings: classical NLP tasks, math reasoning, etc. This was mainly due to the large availability of evaluation datasets in those domains. This is now changing, we see several new types of more realistic benchmarks that necessitate interacting within a given environment (WebArena [25], Webshop [22], OSWorld [21], AppWorld [18]). Learning paradigms that deal with interactivity need to be used.

Games are a convenient tool that enables us to construct scenarios that constructively approximate real scenarios. For example, the complexity of the games (what is observed vs what is learned, the search space for ML) can be iteratively increased or decreased and hence different learning methods can be studied in a more systematic and comparative fashion. Secondly, games, not being samples from real-world interactions but being close approximations of such interaction are a good way to engage human interactors to provide behavioural information and create consistent environments where data-collection (and if needed also data annotation) can be systematically performed.

The working group on Perspectives for Language Learning from Human Interaction produced a classification scheme of current ML approaches for learning from interaction, identifying a number of open questions, including:

- Q4: How can agents learn to have/recognise intentions?
- Q5: What are the tasks/games that can facilitate the acquisition of these skills?

■ **Perceptual Grounding for Embodied Conversational Agents.**

This group built their discussion around the research hypothesis that interactivity plays a major role in human intelligence; interactivity has multiple aspects that we spelled out: 1) interacting with an environment (manipulate objects, act on them), 2) interacting with others through language, and 3) interacting with others while acting in an environment. Through such multimodal embodied experiences humans develop their cognitive intelligence (in other words, understanding the state of affairs in the world) and social intelligence (understanding the mechanisms of interactions).

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3 Overview of Talks

3.1 Assessing Diverse Human Communicative Behaviors in Context

Malihe Alikhani (Northeastern University – Boston, US)

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With the increasing deployment of language technologies to users, the need for a deeper understanding of the impact of natural language processing models on our society and user behaviors has grown. Designing culturally responsible, equitable, and inclusive language technologies that can benefit a diverse population is ever more important. Toward this goal, this brief presentation highlights efforts to address linguistic injustice and promote fairness, with a special focus on sign language technologies. By integrating insights from cognitive science, social sciences, and machine learning, and engaging communities through co-design, we explore pathways for advancing inclusive and innovative language technologies that support diverse communicative needs.

3.2 Overview of games and multimodal platforms useful for conversational AI agents

Raffaella Bernardi (University of Trento, IT), Marc-Alexandre Côté (Microsoft – Montreal, CA), Diego Perez Liebana (Queen Mary University of London, GB), David Schlangen (Universität Potsdam, DE), and Alessandro Suglia (Heriot-Watt University – Edinburgh, GB)

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The literature is flourishing with surveys on this topic taking Large Language Models (LLMs) as the main viewpoint. We summarized the classification reported in these surveys and the future directions they highlight. It becomes clear that Interactive Games, within 3D environments, and multi-agent situations will play a crucial role in the near future. Such games are a good tool to evaluate LLMs and to advance from models to agents, and from virtual environments to real-world situations. From such an overview, it also emerged the importance of paying attention to evaluation methods. In particular, the next challenges are in evaluating reasoning and planning capabilities, as well as the social intelligence of AI systems. After this overview, we focused on reporting the roles LLMs have currently in games, the limitations they are facing, and the ethical issues of using LLMs in games. Furthermore, we reviewed the existing virtual and 3D environments. Finally, we identified open questions focusing on the evaluation methods for aspects of multimodal interaction currently not evaluated, for testing what would be gained by moving to multimodal interactive games and what are the limitations of current games and platforms.

3.3 Modular Interactive Language Models

Marc-Alexandre Côté (Microsoft – Montreal, CA)

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Joint work of Marc-Alexandre Côté, Alessandro Sordoni, Lucas Page-Caccia, Xingdi Yuan

The talk motivated the use of a modular approach for specializing language models (LMs) more efficiently. For instance, Low-Rank Adapters (LoRAs) can be learned independently on diverse set of tasks and later be combined during inference via model weights merging. The second part of the talk motivated the need for more interactivity during LMs training since this how LLMs are being deployed nowadays. Also, a brief overview of TW-Bench was given, showcasing the current SoTA performance of LLMs at playing text-based games.

3.4 The grounded and communication task performance abilities of (embodied/multimodal, conversational) AI agents: SOTA

Julia Hockenmaier (University of Illinois – Urbana-Champaign, US), Raffaella Bernardi (University of Trento, IT), Massimo Poesio (Queen Mary University of London, GB & Utrecht University, NL), Raquel Fernandez, and Sina Zarrieß (Universität Bielefeld, DE)

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Current conversational AI agents have remarkable ability to understand written language and static images. In this first State-of-the-Art presentation, we introduced a distinction between language-only tasks, tasks requiring multi-turn interactions (as in chatbots), vision-language tasks (such as captioning), and embodied tasks (real or simulated). We then surveyed the extensive work devoted to testing models in these different tasks – e.g., for referring expression interpretation and intention recognition (theory of mind). Aspects of grounded interaction covered in some detail included grounding (the distinction between “perceptual grounding” and “meaning grounding” was made) and the extent to which agents are aware of the ambiguity of NL expressions in visual tasks.

3.5 How can multimodality be leveraged for coreference resolution?

Nikolai Ilinykh (University of Gothenburg, SE) and Sharid Loáiciga (University of Gothenburg, SE)

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In the traditional textual coreference resolution task, a model identifies whether two or more linguistic expressions in a text refer to the same entity, object, or event. In a multimodal setting, these “referring expressions” are linked, or grounded, to their corresponding visual counterparts. Our work focuses on the interplay between visual information and coreference, investigating how visual cues can assist models in resolving complex coreferential usages.

Today, we present results from a pilot study evaluating textual coreference tools on multimodal data. We also outline our roadmap for developing a model that integrates visual and textual features to enhance coherence. Our approach addresses traditionally challenging

cases, such as plural pronouns, and involves annotating additional data to capture a broader range of phenomena. We aim to generate text using coreference chains tailored to specific target populations and aimed at mitigating hallucination. Finally, we aim to work with other genres, including dialog.

3.6 Optimization dialogue games

Alexander Koller (Universität des Saarlandes – Saarbrücken, DE)

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Main reference Isidora Jeknic, David Schlangen, Alexander Koller: “A Dialogue Game for Eliciting Balanced Collaboration”, in Proc. of the 25th Annual Meeting of the Special Interest Group on Discourse and Dialogue, SIGDIAL 2024, Kyoto, Japan, September 18 – 20, 2024, pp. 477–489, Association for Computational Linguistics, 2024.

URL <https://doi.org/10.18653/v1/2024.sigdial-1.41>

The talk introduces a new type of dialogue game, in which two players collaborate to jointly solve an NP-hard optimization problem (e.g. two-player Traveling Salesman). These games bring together two lines of my recent work: (a) dialogue games that are designed to elicit balanced role-taking, as in Jeknic et al. 2024, and (b) the use of dressed-up NP-hard problems as test suites for the reasoning capabilities of LLMs. Preliminary results indicate that current LLMs can play *some* optimization games successfully, especially if the graphs are small, but lose track of the dialogue state after a few turns and are unable to solve the problems.

3.7 Introduction to the Dagstuhl Perspectives Workshop 24492: Human-in-the-Loop Learning through Grounded Interaction in Games

Massimo Poesio (Queen Mary University of London, GB & Utrecht University, NL), Raffaella Bernardi (University of Trento, IT), Julia Hockenmaier (University of Illinois – Urbana-Champaign, US), and Udo Kruschwitz (Universität Regensburg, DE)

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In recent years we have witnessed an explosion of research in three converging directions. The first is a decisive move in Artificial Intelligence (AI) towards human-centered intelligence and AI models that can learn through interaction and are able to act as intelligent assistants. The second is equally substantial process in conversational agents and dialogue systems on grounded interaction – developing models aware of the situation in which they operate (a physical world in the case of robots, a virtual world in the case of artificial agents) and able to, e.g., understand / produce references to this situation. The third is the appearance of virtual world games in which conversational agents impersonating characters can learn to perform tasks, or improve their communicative ability, by interacting with human players in platforms such as Minecraft. The objective of Dagstuhl Perspectives Workshop 24492 was to review progress in these directions, and map the most promising future research directions. In this introductory presentation, I summarized motivations and objectives of the workshop, and presented the preliminary schedule.

3.8 The “Learning in Interaction” Challenge

David Schlangen (Universität Potsdam, DE)

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Joint work of David Schlangen, Raffaella Bernardi, Alexander Koller, Alessandro Suglia
Main reference Kranti Chalamalasetti, Jana Götze, Sherzod Hakimov, Brielen Madureira, Philipp Sadler, David Schlangen: “clembench: Using Game Play to Evaluate Chat-Optimized Language Models as Conversational Agents”, in Proc. of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023, pp. 11174–11219, Association for Computational Linguistics, 2023.
URL <https://doi.org/10.18653/V1/2023.EMNLP-MAIN.689>

In this talk, I presented an overview of a planned community challenge that is aimed as fostering research on “learning in interaction”. I started by presenting a motivation for why one might want to look into learning in interaction (as opposed to learning from observation): Ecological plausibility, strong learning signal. I then talked about some challenges in realising this idea, stemming from the sample inefficiency of our current learning mechanisms, which make it difficult to set up “human-in-the-loop” learning. The challenge starts from the observation that the situation has changed somewhat: We now have available to good (albeit not perfect) “teacher simulators” that do not tire: Large Language Models. The idea of the challenge is that so-called “frontier models” (the current largest and most competent ones) can be used to guide the learning process of smaller models, leading to a relative improvement that is larger than that achievable by other learning methods on the same amount of data. The challenge will build on the clembench project, to provide participants with an environment in which they can bring their learner models in interaction with stronger teacher models, in a range of conversational games.

3.9 Current Approaches to Human (and Artificial Agent)-in-the-Loop Learning for AI Agents

Alessandro Suglia (Heriot-Watt University – Edinburgh, GB), Malihe Alikhani (Northeastern University – Boston, US), Marc-Alexandre Côté (Microsoft – Montreal, CA), Alexander Koller (Universität des Saarlandes – Saarbrücken, DE), David Schlangen (Universität Potsdam, DE), Edwin Simpson (University of Bristol, GB), and Alane Suhr (University of California – Berkeley, US)

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The most recent breakthroughs in large language models have been possible thanks to the development of Transformer-based models that can be trained at scale using self-supervised learning objectives, which are dependent upon access to internet-scale datasets. Current models learn from these datasets assuming that they are “observers” – lacking any form of interaction. In this working group, we discussed the importance of interaction with the world and with other agents as an important means for learning competent artificial agents that can solve a variety of real-world tasks that involve coordination and planning of actions. We first provided a concrete definition for “interaction” and what can be learned through interaction. We presented a list of prior works for learning in interaction ranging from the pre-LLM era to cutting-edge research. Interestingly, most of them leverage language in one form or another to provide/guide the learning process. As a result of our investigation, encompassing

recent literature on reinforcement learning, we found that current learning approaches are predominantly focused on single-turn interaction and that very few deal with the intricacies of real-world interactions required to complete complex real-world tasks. This includes an inability to quantify uncertainty in beliefs relating to the environment or interaction partners or to request clarifications. Some rely on human preference data collected off-policy – not through interaction with the learner –, which leads to models that struggle to reflect human preferences, particularly over many turns of interaction. In essence, we argue that future work should develop learning methods that are able to leverage language use in interaction to acquire more relevant signals for learning more robust agent policies.

3.10 The Cooperative Testing Initiative

Alane Suhr (University of California – Berkeley, US)

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Joint work of Alane Suhr, Alexander Koller

Real human conversation is messy and flexible, and its dynamics are influenced by interaction design. In this talk, I describe a new corpus of goal-directed conversations between human players of the cooperative 3D puzzle game, Portal 2, and outline a number of complex phenomena that arise in real human conversations in the corpus.

3.11 Exploring new variations of reference games

Sina Zarrieß (Universität Bielefeld, DE) and Albert Gatt (Utrecht University, NL)

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Joint work of Simeon Junker, Sina Zarrieß, Albert Gatt

This talk discussed reference games as a tool to study notions of context and context effects on language use in grounded interaction. As our starting point, we introduced classic reference games where the speaker’s task is to refer to a target object, in the context of a set of distractor objects. As an example, we showed a reference game from a basic color domain, where even recent multimodal LLMs do not resolve references to simple color patches correctly when the referring expressions are highly context-sensitive. We compared such synthetic toy settings to reference games where more realistic scenes, e.g. images, are available. These contexts are more akin to scenes that humans act in in their everyday lives. We argue that these scenes broaden the types of context and context effects that can be studied. Based on [1], we show that scene context can be relevant for recognizing target objects in referring expression models, but also show that current models still seem to lack a holistic scene and situation understanding. We discuss examples of ongoing data collections, where these ideas are extended further to test the effects of scene-level expectations and violations on reference and the effects of scene context on conceptualization.

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4 Working groups

4.1 Complex Interactions

Elisabeth André (Universität Augsburg, DE), Jonathan Ginzburg (University Paris-Diderot, FR), Alexander Koller (Universität des Saarlandes – Saarbrücken, DE), Matthew Purver (Queen Mary University of London, GB), Edwin Simpson (University of Bristol, GB), and Alane Suhr (University of California – Berkeley, US)

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The Complex Interaction working group identified complexities of language-based interactions that arise from interaction design, including incentive mechanisms and participant affordances (in action and communication). They enumerated a number of such complex phenomena, and for each, define the phenomenon, provide some background on historical approaches to addressing the phenomenon, the state of current models in handling the phenomenon, and any recent approaches addressing them. For each of these, they also described what one might learn in interaction when the phenomenon occurs. Finally, they provided recommendations on different philosophies in designing systems that are able to manage such complexities in interactions with humans.

4.2 Perceptual Grounding and Embodiment

Raffaella Bernardi (University of Trento, IT), Ryuichiro Higashinaka (Nagoya University, JP), Julia Hockenmaier (University of Illinois – Urbana-Champaign, US), Sharid Loáiciga (University of Gothenburg, SE), Catharine Oertel (TU Delft, NL), and Carina Silberer (Universität Stuttgart, DE)

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Discussed Problems

This group discussed a number of topics, from perceptual grounding to theory of mind. Much of the discussion was centered around the notion of agents (or humans) interacting with each other or the environment.

We distinguished between different levels of interactivity: 1) interacting with an environment (manipulating objects to act on them), 2) observing or interacting with others (who may operate in an environment) purely through language, and 3) interacting with others while acting in an environment, each requiring specific capabilities. Inspired by the cooperative game *Overcooked*, we tried to map these levels to the cooking domain as 1) a single cook, 2) a travel cookbook writer who observes how locals prepare food, and 3) a member of a kitchen team.

Humans develop their cognitive intelligence (in other words, understanding the state of affairs in the world) and social intelligence (understanding the mechanisms of interactions) through such multimodal embodied experiences. We therefore began to spell out which capabilities embodied agents would require to perform tasks that involve these different degrees of interactivity, and how such capabilities could be evaluated in artificial agents.

Possible Approaches

The WG proposed to use existing games in virtual environments. We identify different dimensions that should be studied: a) levels of interactivity from observing other agents playing the game to being a player of the game; b) complexity of the games; c) complexity of the perceptual and social input; d) complexity of the interactions.

Conclusions

Important milestones set in the ‘50 by the AI community have been reached thanks to amazing progress in Machine Learning, Computer Vision, and Natural Language Processing. The new challenges are now on how to push SOTA AI models to face the challenges posed by the plethora of disciplines that have been part of AI since its inception. Of this, the WG focused on Cognitive Science and Linguistics, highlighting the research lines on common ground, perceptual grounding and reference, embodied cognition, and social science.

4.3 Perspectives for Language Learning from Human Interaction

Alessandro Suglia (Heriot-Watt University – Edinburgh, GB), Malihe Alikhani (Northeastern University – Boston, US), Marc-Alexandre Côté (Microsoft – Montreal, CA), Simon Dobnik (University of Gothenburg, SE), Haishuo Fang (TU Darmstadt, DE), David Schlangen (Universität Potsdam, DE), and Andrew Zhu (University of Pennsylvania – Philadelphia, US)

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Discussed Problems

In this working group, we discussed the problem of learning language through interaction. We structured the discussion around the language development phases of an infant. Without going in too much detail about the process of (human) first language acquisition, we can note one interesting difference in the way language “acquisition” in machines is set up. For the human infant, the need (and want) to interact and communicate intentions comes long before the sophisticated ways of expressing intentions through language are acquired. Our current machine learning methods do not make this distinction, and if dealing with intentions is acquired at all (which is doubtful), it is acquired through the acquisition of the statistical patterns of language. We also note the crucial role that interaction plays in human language acquisition (be it interaction that the child is involved in or observed interaction), with other modes of language use (e.g., written language) only being acquired through formal training (and not universally) – writing is a technology.

Possible Approaches

Based on this investigation, we have surveyed the literature on reinforcement learning and derived three possible training regimes that can be used for learning to interact with the world and with other agents: A) on-policy learning which is akin to the classic on-policy RL learning where the policy interacting with the environment is the same one that is updated; B) off-policy learning which is when the interaction data that the agent uses for learning are potentially generated by different instantiations of the agent or by other agents; C) offline learning: this is a form of learning where the agent acts merely as an “observer” of interaction data which it can use to learn from.

Based on this categorisation, we have surveyed the literature and found that most of the methods currently used for training LLMs typically fell under category C treating them as “observers” in the learning process. Therefore, we have highlighted promising training regimes for learning in interaction that take advantage of heuristic-based search strategies (e.g., Monte-Carlo Tree Search) combined with the power of neural networks and large language models.

Conclusions

This working group critically analysed the literature on learning methods for developing artificial agents using LLMs. We argue that in order to have more capable models, it is essential to be able to develop agents that can learn to learn in interaction – going beyond current state-of-the-art methods that treat them as observers of interactional data. Thanks to our discussion on the topic, we were able to find important gaps and pose important questions that will help researchers design and implement more capable artificial agents able to learn through interaction with the world and with other agents.

4.4 Game Design for Grounded Interaction

Sina Zarriß (Universität Bielefeld, DE), Nikolai Ilinykh (University of Gothenburg, SE), Prashant Jayannavar (University of Illinois – Urbana-Champaign, US), Udo Kruschwitz (Universität Regensburg, DE), Diego Perez Liebana (Queen Mary University of London, GB), and Massimo Poesio (Queen Mary University of London, GB & Utrecht University, NL)

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Advances in language modeling have led to an explosion of interest in testing agents in games using game platforms and benchmarks for grounded interaction. Existing games and platforms are extremely diverse in their setting, goals, and complexities, and they are being developed in different subfields of AI, NLP, and computational linguistics. Furthermore, within these subfields, games are designed for different purposes and, to some extent, classified using different taxonomies.

The working group on Game Design for Grounded Interaction identified four key research questions arising from this explosion and diversity of games that we suggest should be explored in future research:

- Q1: How can games and game benchmarks be designed more systematically, such that they lead to a deeper understanding of games and the skills that games are testing? How do we generalize skills and agents’ abilities across games?
 - Q2: How important it is to maximizing human engagement and fun in the design of these games?
 - Q3: What role does the complexity of the game is? And how do we measure it?
 - Q4: How do we evaluate agents within and across games? In particular, how do we evaluate whether the skills trained / tested with a game transfer to real world applications?
- Regarding Q1, the WG identified as a promising research direction to develop inventories of skills or taxonomies that make it easier to compare games and agents’ skills across different games, and that allow to disentangle agents skills at an intermediate level.

There has been little investigation into how fun and engagement impact the effectiveness of game designs and presence of different linguistic phenomena in grounded interactions (Q2). While it is less clear whether fun and engagement are important for game design with agent-in-the-loop approach (e.g., agent-led tasks), these two aspects are very important for games that incorporate a human-in-the-loop approach. The WG recommended that developers of grounded interaction games focus more on evaluating the connection between the game engagement level and the intended game outcome. It is also important to involve professional game designers in the process of game world building and to continuously collect and integrate feedback from players.

Although many grounded interaction games / tasks in which conversational agents occur, or that are used to test / train them, are very simple in design and only involve simple forms of interaction others are very complex. An important cluster of questions that need researching is the extent to which the complexity of a grounded interaction game affects its usefulness for the purposes of testing and evaluation (Q3). Such questions include:

1. How complex a game needs to be to be useful for testing / training?
2. Is there a link between complexity and transferability?

The WG's recommendations are that

- More research should be carried out questions 1 and 2.
- Developers should be more explicit / aware about the complexity in their game and how that may affect its usefulness.

Finally, regarding Q4 – evaluation often prioritizes performance of an agent in the game itself (**intrinsic evaluation**), while placing less emphasis on how in-game performance translates to downstream applications, other games, or real-world scenarios (**extrinsic evaluation**). Both intrinsic and extrinsic evaluations are complementary and equally important.

Evaluation, much like game design, benefits greatly from a **multidisciplinary approach** to ensure robustness and reliability. For instance, the outcomes of human evaluation studies can be influenced by factors such as the personality traits of the recruited participants. Incorporating expertise from fields like psychology during the design of evaluation methodologies can help mitigate potential biases or adverse effects arising from such factors. Similarly, expertise from HCI can play a key role in ensuring that the UX and UI elements needed for human evaluation are designed correctly.

Transferability is an important dimension of evaluation that warrants special emphasis. It examines the extent to which skills learned in one game can be effectively applied to another experimental setting, including both other games and real-world environments. This has implications for the development of generalizable, adaptable learning agents.

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